



Transfer Learning & Finetuning

73.69 - LARGE LANGUAGE MODELS - 2026

Ing. Eugenia Piñeiro, Ing. Marina Fuster

06

Reasoning Models

¿Qué significa que un modelo
de lenguaje *razona*?

¿Cómo definir un *Reasoning Model*?

Son aquellos modelos que **generan tokens adicionales** para resolver un problema.

Dichos tokens adicionales se corresponden con la generación de **pasos** intermedios (verificación, planeamiento, etc.) que ayudan a resolver problemas de mayor complejidad.

¿Cómo definir un *Reasoning Model*?

On a 2016 Toyota Prius the front right door is removed, stripped down and repaired, and the window regulator is replaced. Given the operations performed, what must be replaced to adhere to manufacturer methods?

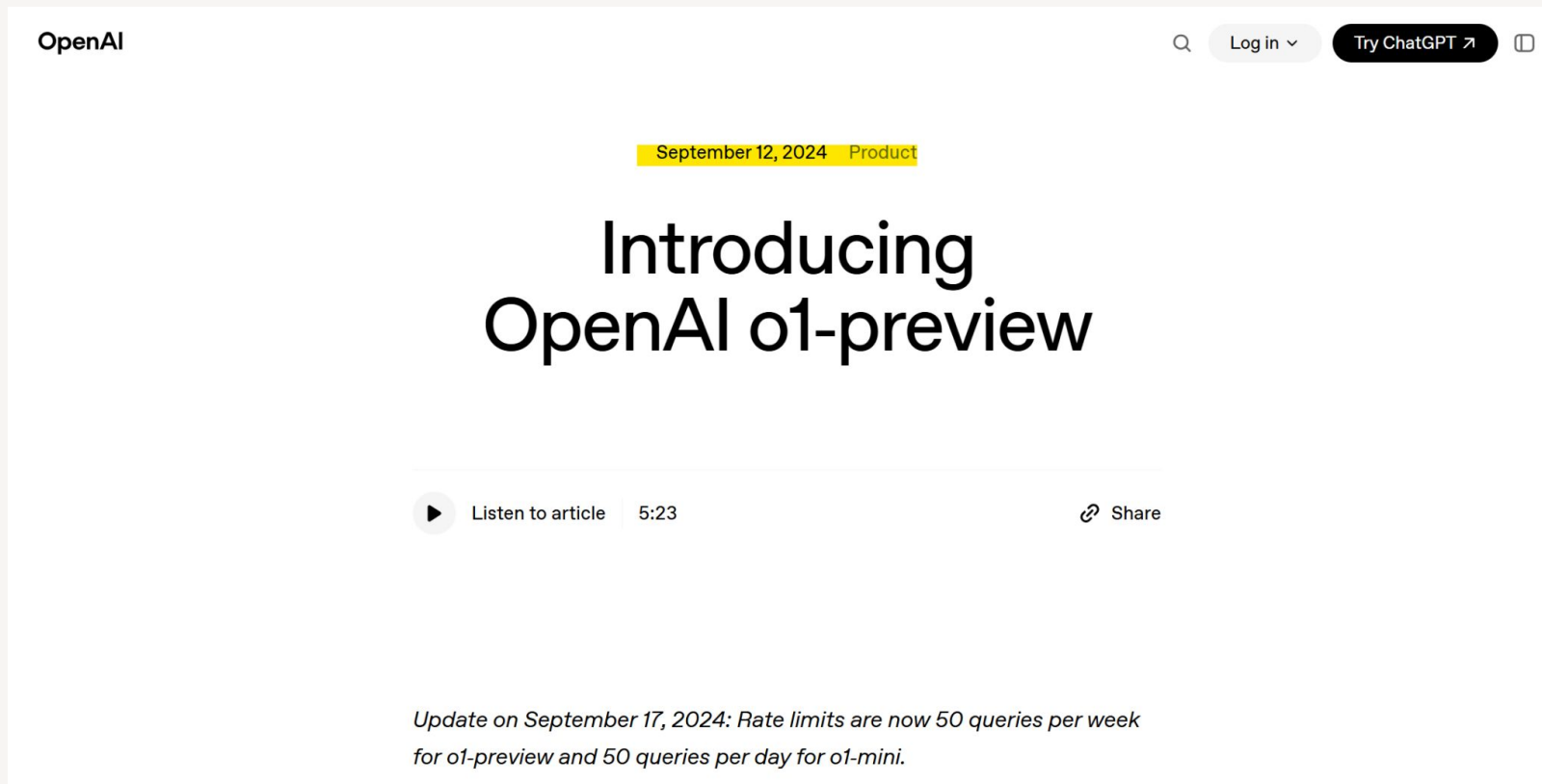


Thought for 6 seconds ▾

When performing the removal, disassembly, repair, and reassembly of the front right door on a 2016 Toyota Prius—including the replacement of the window regulator—Toyota's manufacturer repair methods specify that certain components must be replaced to ensure proper function and safety. Specifically, you must replace:

1. **Front Door Service Hole Cover (Vapor Barrier):** This is a seal that prevents moisture and debris from entering the vehicle's interior. Once removed, it cannot be reused due to potential damage to the adhesive and seal integrity.
2. **Single-Use Fasteners and Clips:** Any non-reusable clips, retainers, or self-locking nuts that were removed during disassembly must be replaced. These components are designed for one-time use to maintain proper clamping force and alignment.

¿Cómo definir un *Reasoning Model*?



The screenshot shows the OpenAI website header with the logo, a search icon, a 'Log in' button, and a 'Try ChatGPT' button. Below the header, a date and category tag 'September 12, 2024 Product' is displayed. The main heading reads 'Introducing OpenAI o1-preview'. A media player section includes a play button icon, the text 'Listen to article', a duration of '5:23', and a 'Share' link. At the bottom, an update note states: 'Update on September 17, 2024: Rate limits are now 50 queries per week for o1-preview and 50 queries per day for o1-mini.'

OpenAI

September 12, 2024 Product

Introducing OpenAI o1-preview

Listen to article 5:23 Share

Update on September 17, 2024: Rate limits are now 50 queries per week for o1-preview and 50 queries per day for o1-mini.

Inference Time

Inference Time

CoT prompting

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

Inference Time

Training Time



CoT prompting

Inference Time

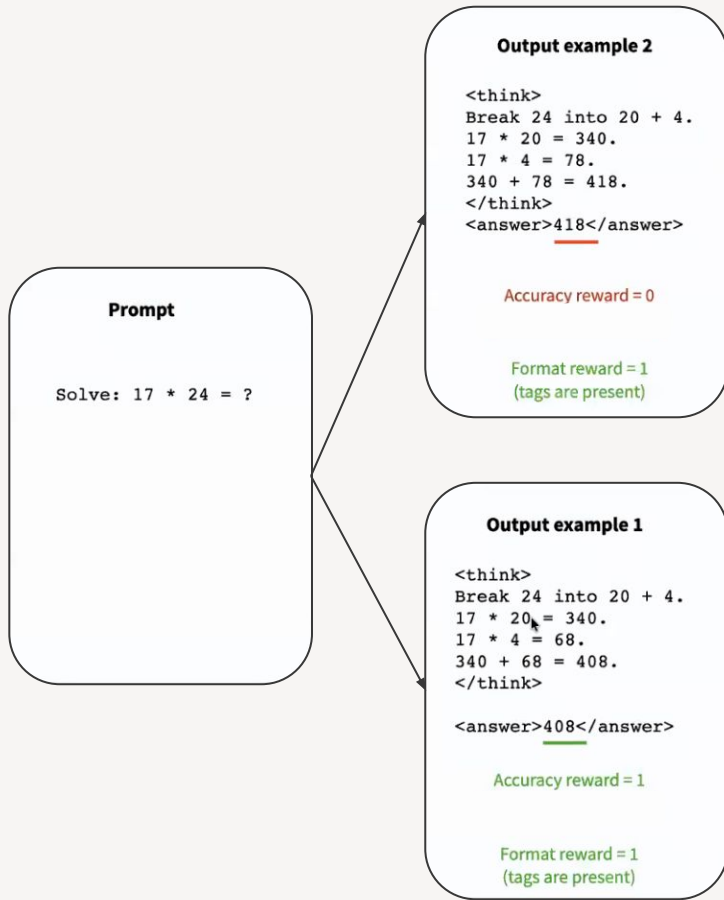


CoT prompting

Training Time



RLVR



1. Inference Time Scaling

The idea is straightforward. If we are willing to spend a bit more compute, and more time at inference time (when we use the model to generate text), we can get the model to produce better answers.

[Sebastian Raschka, PhD](#)

1. Inference Time Scaling

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

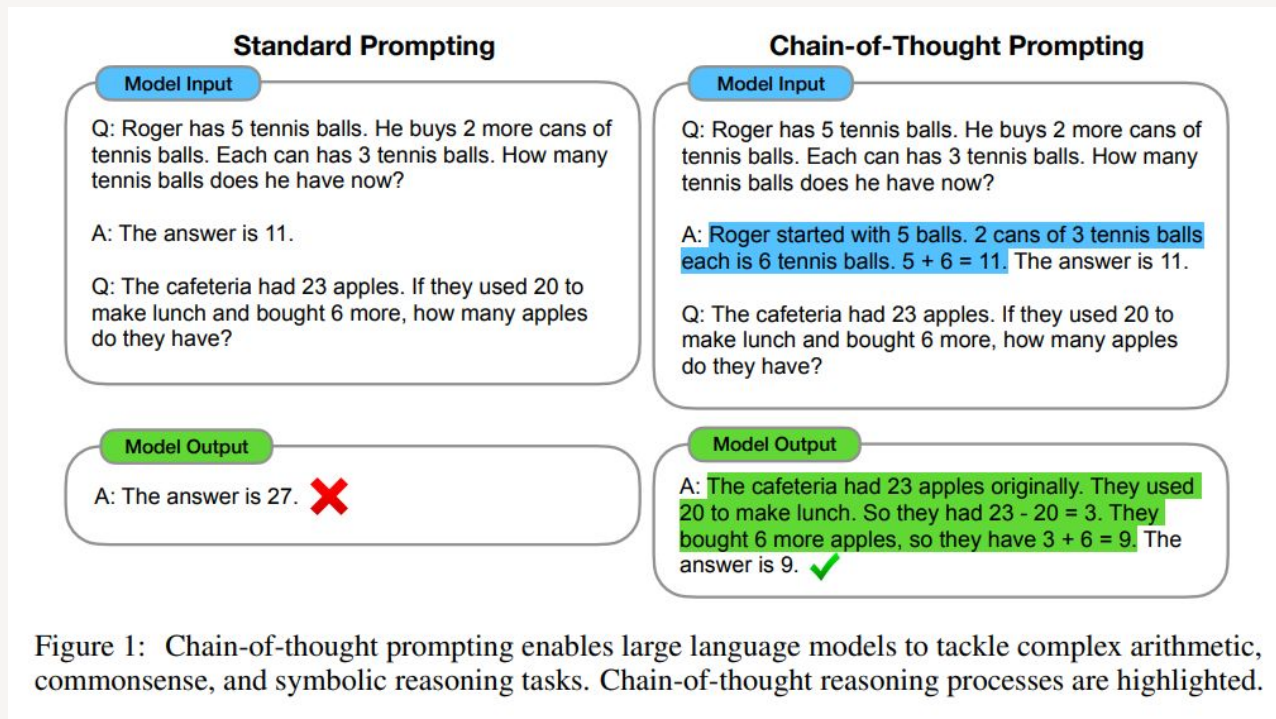
Jason Wei Xuezhi Wang Dale Schuurmans Maarten Bosma
Brian Ichter Fei Xia Ed H. Chi Quoc V. Le Denny Zhou

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Abstract

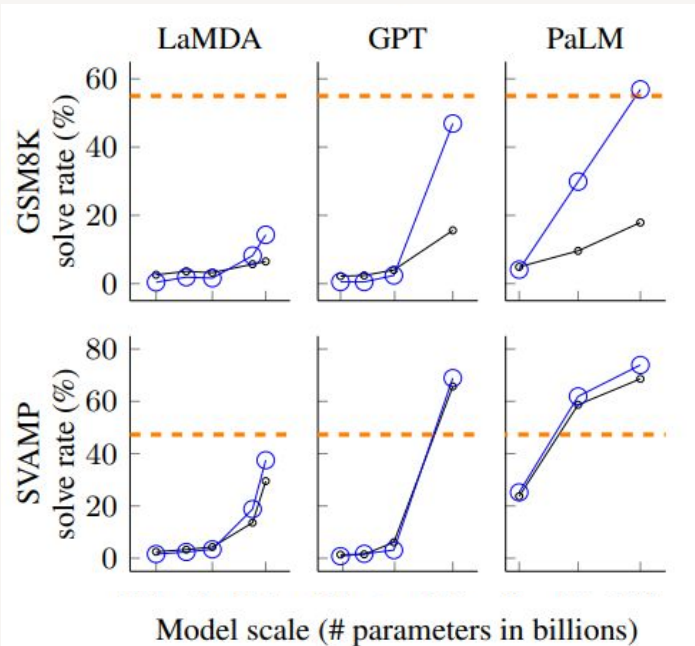
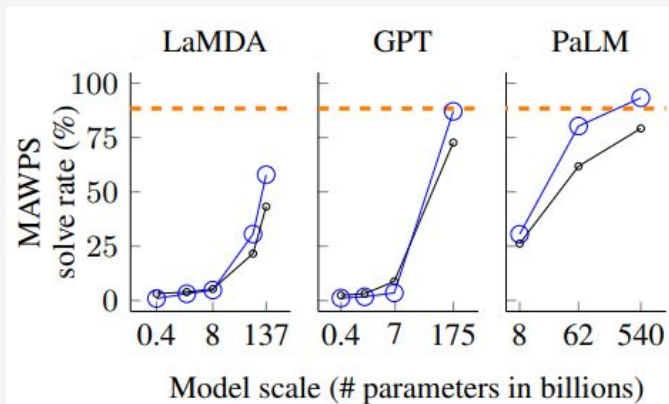
We explore how generating a *chain of thought*—a series of intermediate reasoning steps—significantly improves the ability of large language models to perform complex reasoning. In particular, we show how such reasoning abilities emerge naturally in sufficiently large language models via a simple method called *chain-of-*

1. Inference Time Scaling



1. Inference Time Scaling

- Standard prompting
- Chain-of-thought prompting
- - - Prior supervised best



1. Inference Time Scaling

- CoT prompting
- **Self-consistency**
 - **sampling!**
- Self-refine
- Agentic Tool Use (model scaffolding)

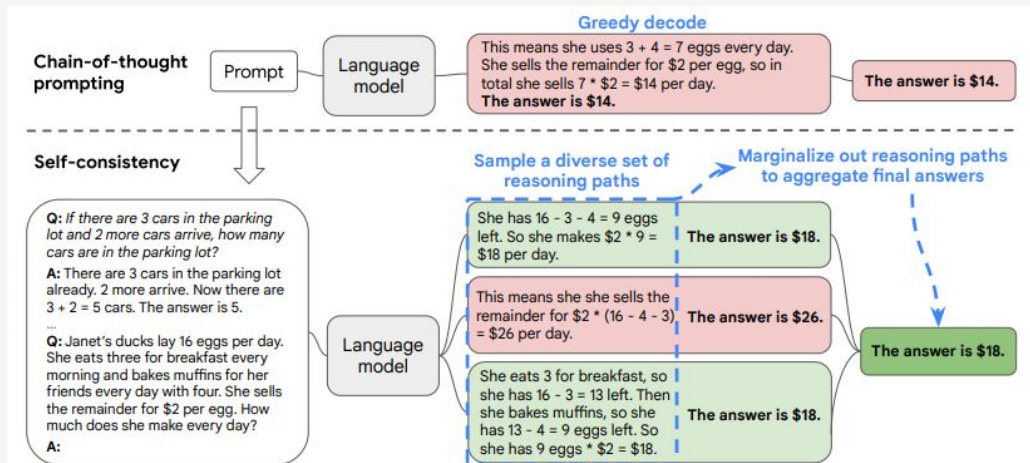
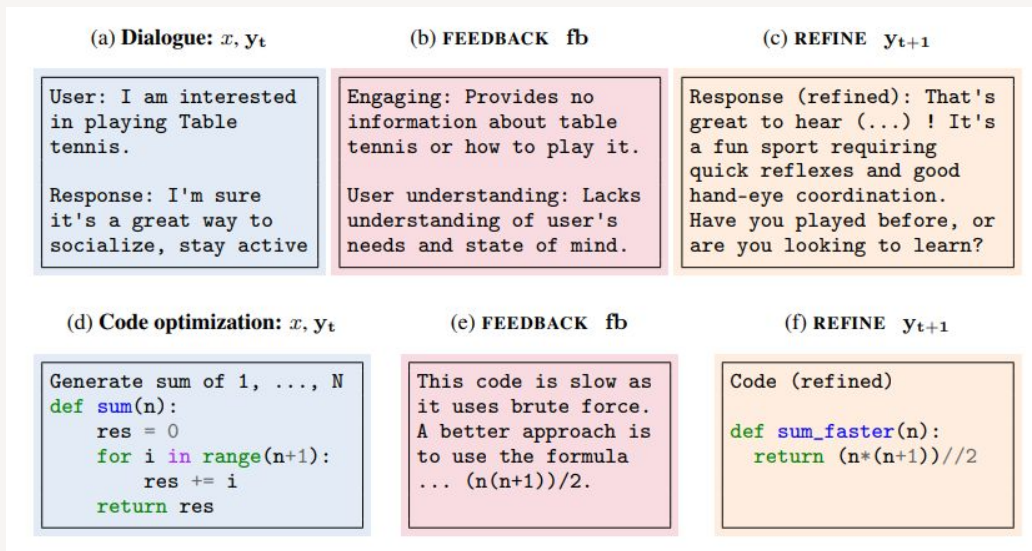


Figure 1: The self-consistency method contains three steps: (1) prompt a language model using chain-of-thought (CoT) prompting; (2) replace the “greedy decode” in CoT prompting by sampling from the language model’s decoder to generate a diverse set of reasoning paths; and (3) marginalize out the reasoning paths and aggregate by choosing the most consistent answer in the final answer set.

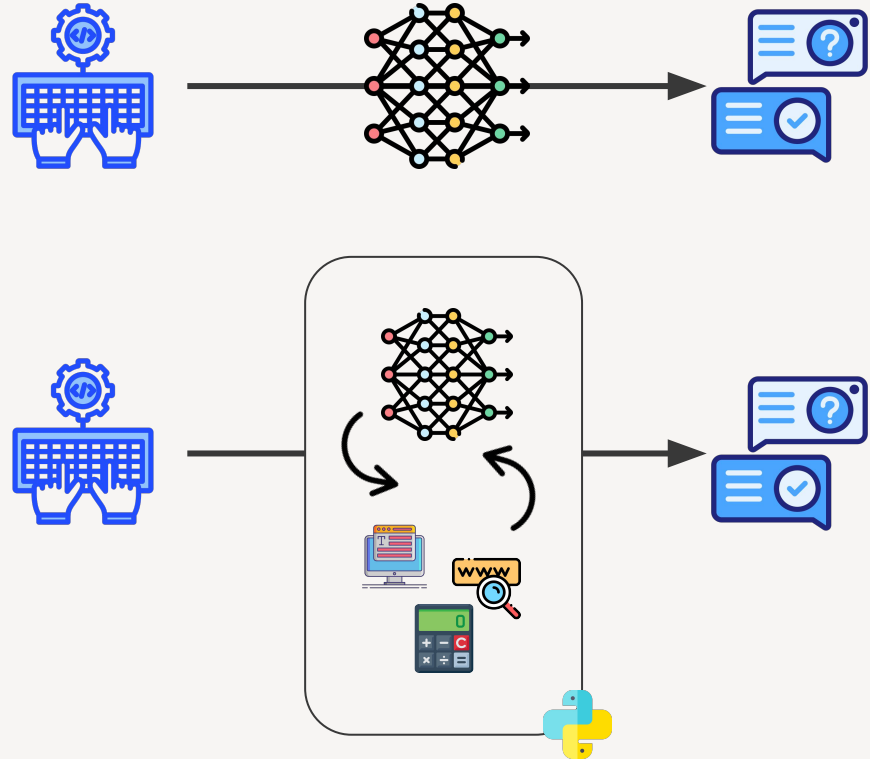
1. Inference Time Scaling

- CoT prompting
- Self-consistency
 - sampling!
- **Self-refine**
- Agentic Tool Use (model scaffolding)



1. Inference Time Scaling

- CoT prompting
- Self-consistency
 - sampling!
- Self-refine
- **Agentic Tool Use (model scaffolding)**



2. Reinforcement Learning

Step 1

Collect demonstration data, and train a supervised policy.

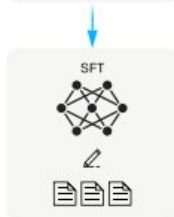
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.



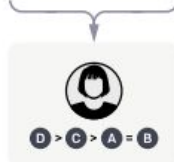
Step 2

Collect comparison data, and train a reward model.

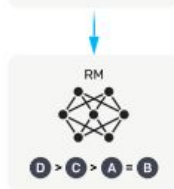
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



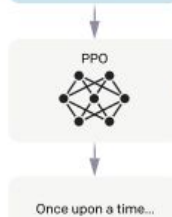
Step 3

Optimize a policy against the reward model using reinforcement learning.

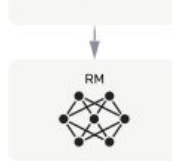
A new prompt is sampled from the dataset.



The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



2. Reinforcement Learning

RLHF



4 modelos!

PPO (ejemplo simplificado)

Prompt: "do pigs fly?"

Answer: "yes"

$$J(\theta) = \mathbf{R}(\text{do pigs fly? yes}) \cdot \log \pi(\text{yes} \mid \text{do pigs fly?})$$

$$J(\theta) = \mathbf{-2.8} \cdot \log 0.7$$

$$J(\theta) = 0.999$$

Reemplazado por **advantage**

$$\pi_{\text{new}} / \pi_{\text{ref}}$$

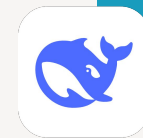
Clipped

Backprop (gradient ascent)

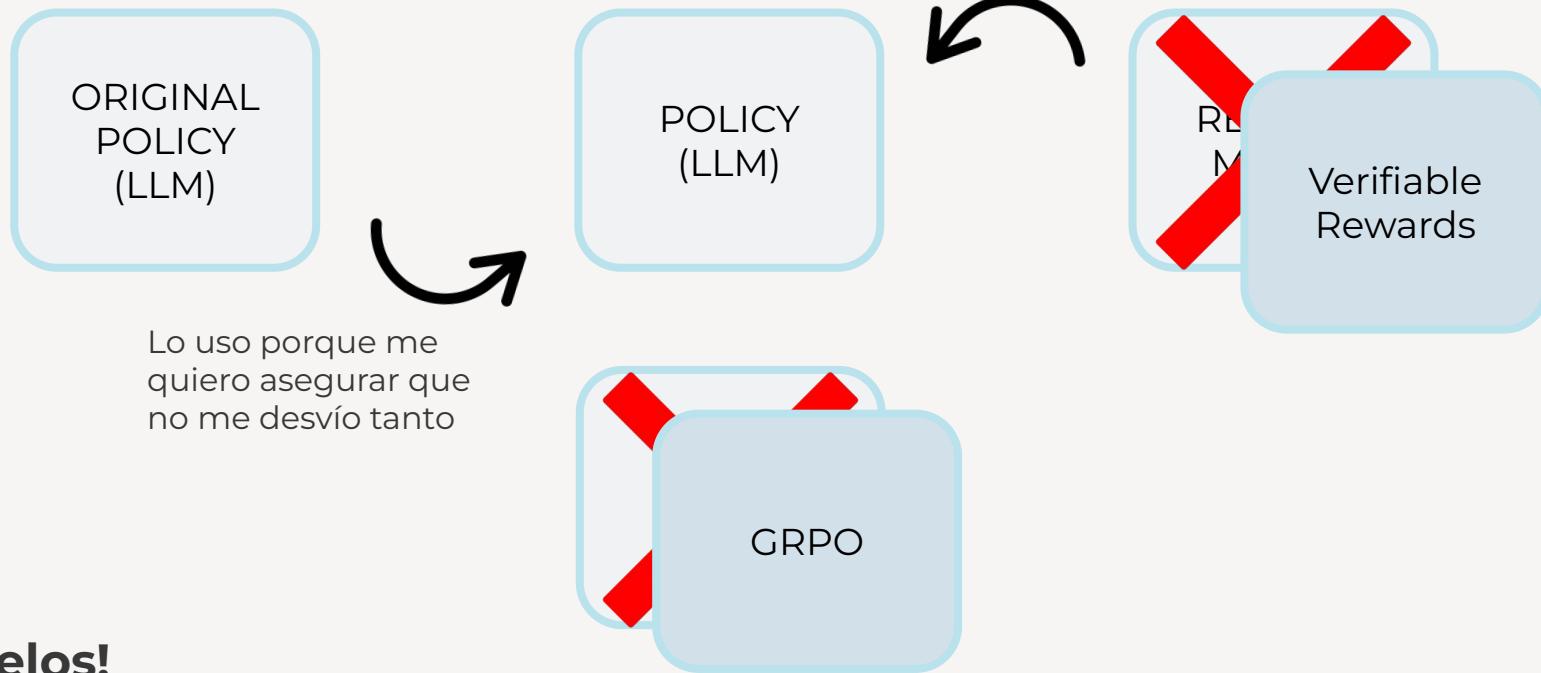
$$J_{\text{final}}(\theta) = J(\theta) + \text{KL penalty}$$

De la clase
pasada!

2. Reinforcement Learning



RLVR



2 modelos!

2. Reinforcement Learning

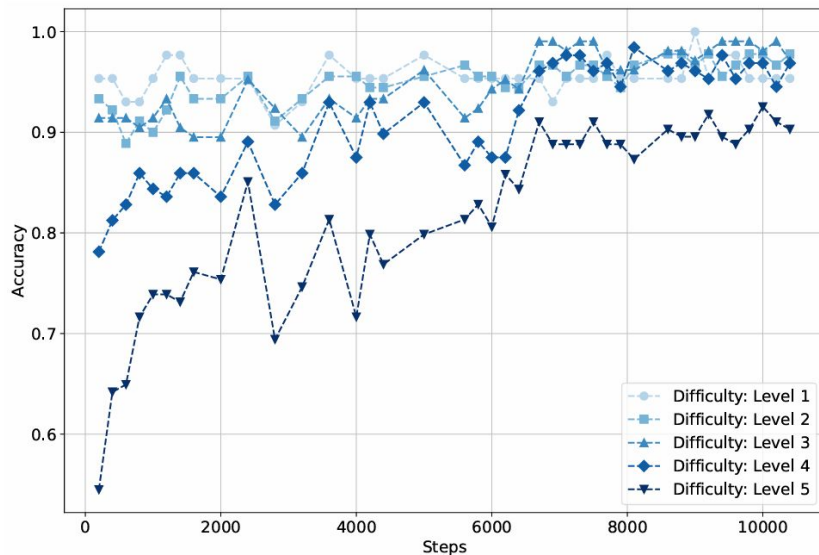
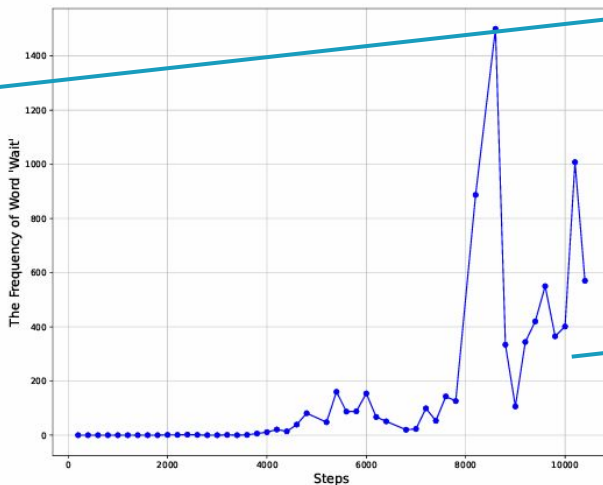
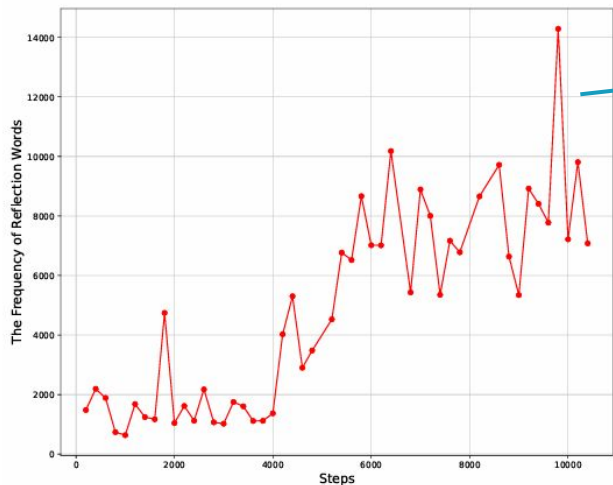


Figure 8 | Performance of DeepSeek-R1-Zero on problems with varying difficulty levels in the MATH dataset.

2. Reinforcement Learning



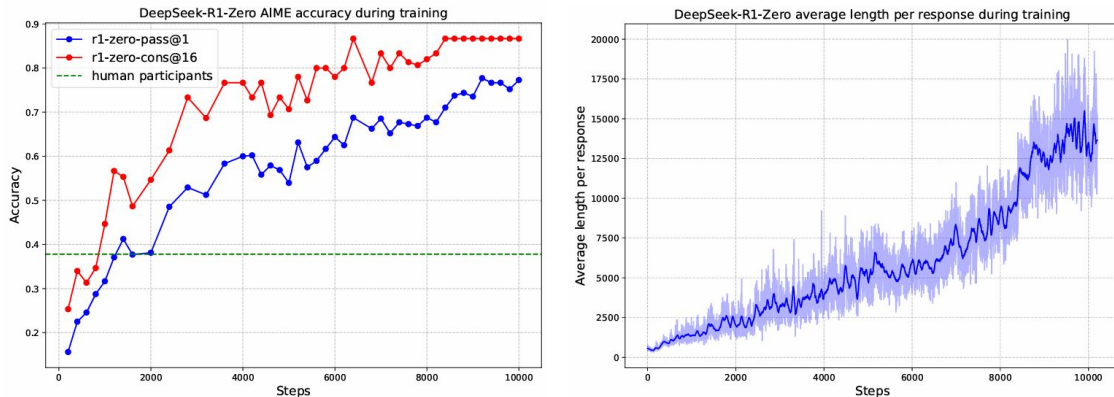
Puede ser
indicativo de
"reflexión"

Puede ser
indicativo de
"auto
corrección"

Figure 9 | Evolution of reasoning behaviors during training. (a) Frequency of representative reflective words during the training process; (b) Specific occurrence patterns of the word "wait" throughout the training process.

2. Reinforcement Learning

RLVR

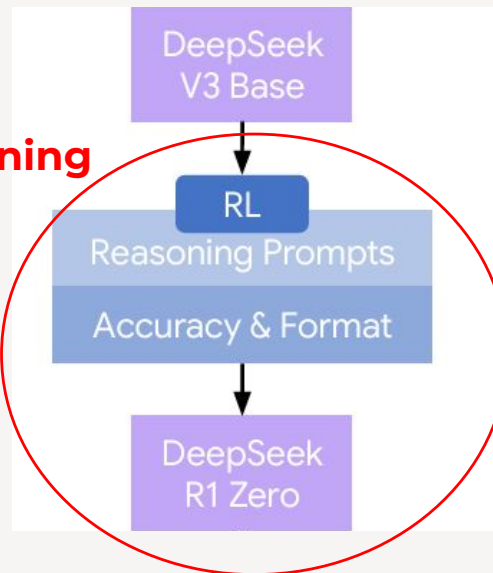


Comienza a
“utilizar más
computo”

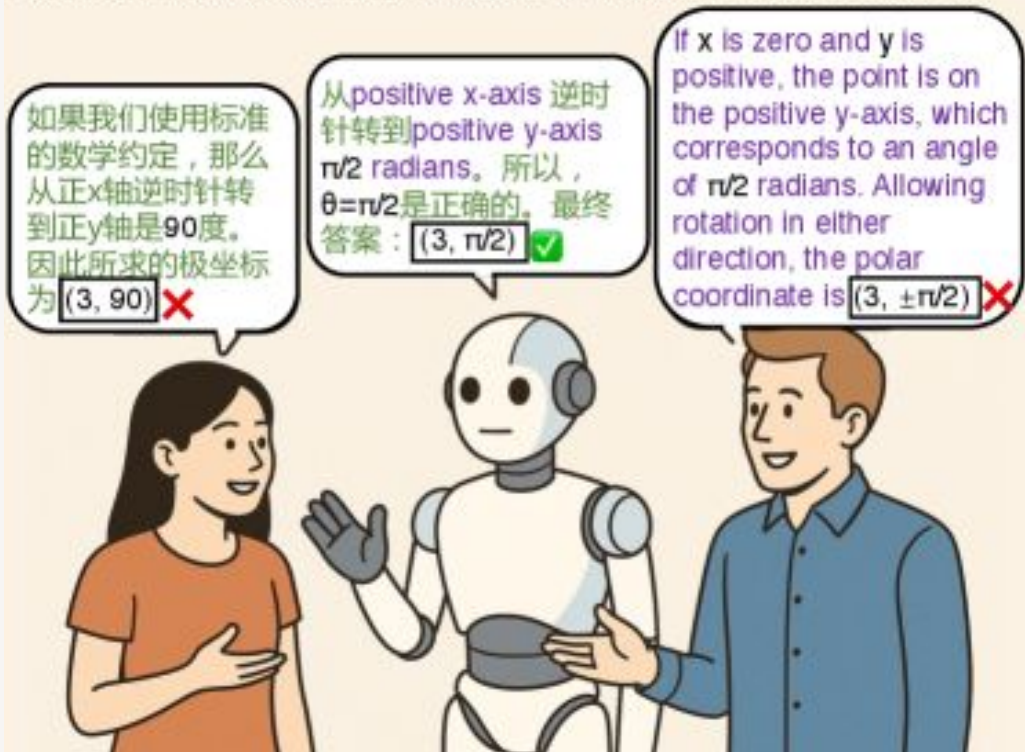
Figure 1 | (a) AIME accuracy of DeepSeek-R1-Zero during training. AIME takes a mathematical problem as input and a number as output, illustrated in Table 32. Pass@1 and Cons@16 are described in Supplementary D.1. The baseline is the average score achieved by human participants in the AIME competition. (b) The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time. Note that a training step refers to a single policy update operation.

¿Qué problemas podría traer
entrenar “razonamiento” de
esta forma?

Reasoning



Q: Convert the point (0,3) in rectangular coordinates to polar coordinates.



[The Impact of Language Mixing on Bilingual LLM Reasoning](#)

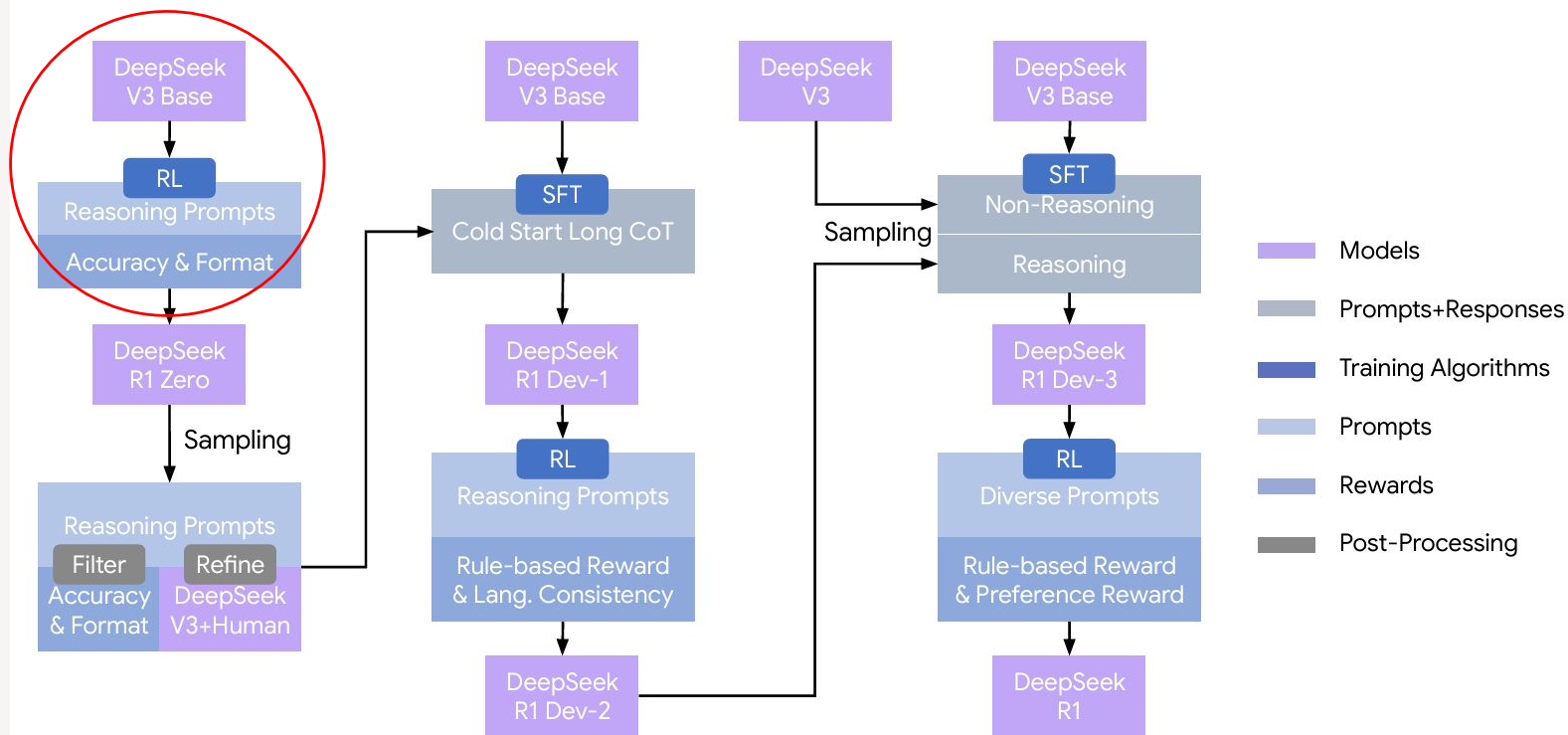


Figure 2 | The multi-stage pipeline of DeepSeek-R1. A detailed background on DeepSeek-V3 Base and DeepSeek-V3 is provided in Supplementary [A.1](#). The models DeepSeek-R1 Dev1, Dev2, and Dev3 represent intermediate checkpoints within this pipeline.

**Inference Time
Scaling**

**Reinforcement
Learning**

SFT + RL

**Knowledge
distillation
(SFT | KD)**

03

LOS
CONDIMENTOS,
UN RESUMEN

Modelo base

Corpus, Self-Supervised
Learning, Scaling Laws

Asistente

Supervised Fine Tuning

Seguridad***

RL or SFT (principalmente RL)
como estrategias de fine
tuning (RLHF, DPO)

Razonamiento

RL como estrategia de fine
tuning y prompting como
estrategia de test-time
inference



Klaas  

@forgebitz



i changed all our "loading..." states to "thinking.."

we are an agentic AI startup now



LLMs y desarrollo en producción

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Etapas del desarrollo de un LLM (repaso)

PROCESOS INVOLUCRADOS

CONSTRUCCIÓN

- Recopilación de datos
- Limpieza de datos y preparación
- Selección de arquitectura e hiperparámetros

PRE-ENTRENAMIENTO

- Proceso computacional para ajustar parámetros (training)
- Evaluación de modelo

FINE-TUNING

- Ajuste hacia una tarea específica
- Añadir capas de seguridad o de características específicas sobre el comportamiento del modelo

PRE
ENTRENAMIENTO

POST
ENTRENAMIENTO

PROCESOS INVOLUCRADOS

CONSTRUCCIÓN

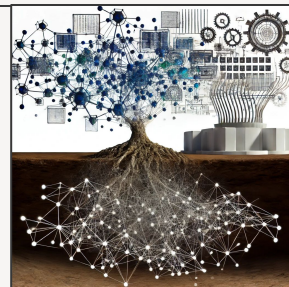
- Recopilación de datos
- Limpieza de datos y
- Selección de arquitectura + Dataset



PRE-ENTRENAMIENTO

- Proceso computacional para ajustar parámetros
- Evaluación de modelo

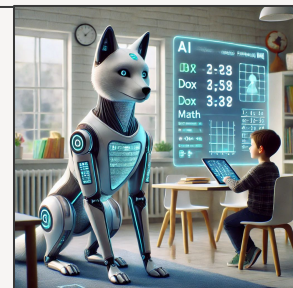
Producto:
Foundational
Model



FINE-TUNING

- Ajuste hacia una tarea específica
- Añadir capas de seguridad o de características al comportamiento del modelo

Producto:
Asistente/
Clasificador/
Generador/
Juez/
etc.

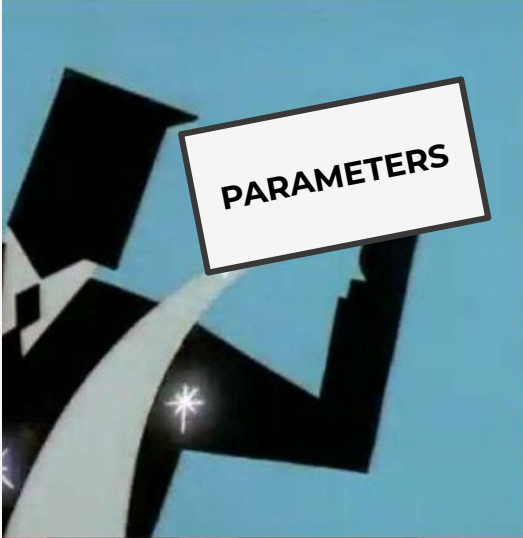


Avances/Producción

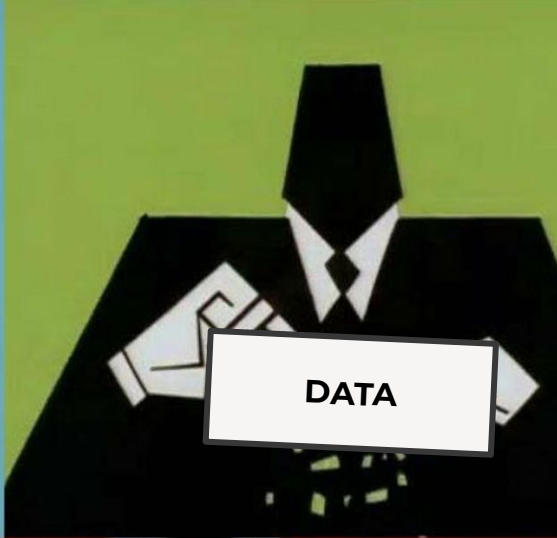
Una (muy) selectiva lista de prácticas:

- Scaling Laws
- Mixture of Experts
- RoPE
- Construcción del corpus
- LoRA(*)
- Quantization(*)

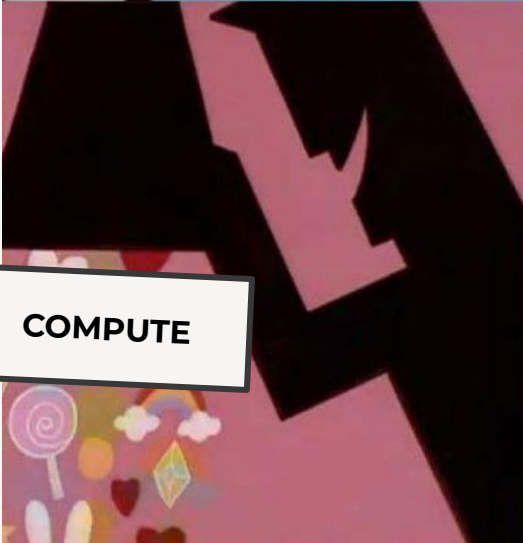
Scaling Laws



PARAMETERS



DATA



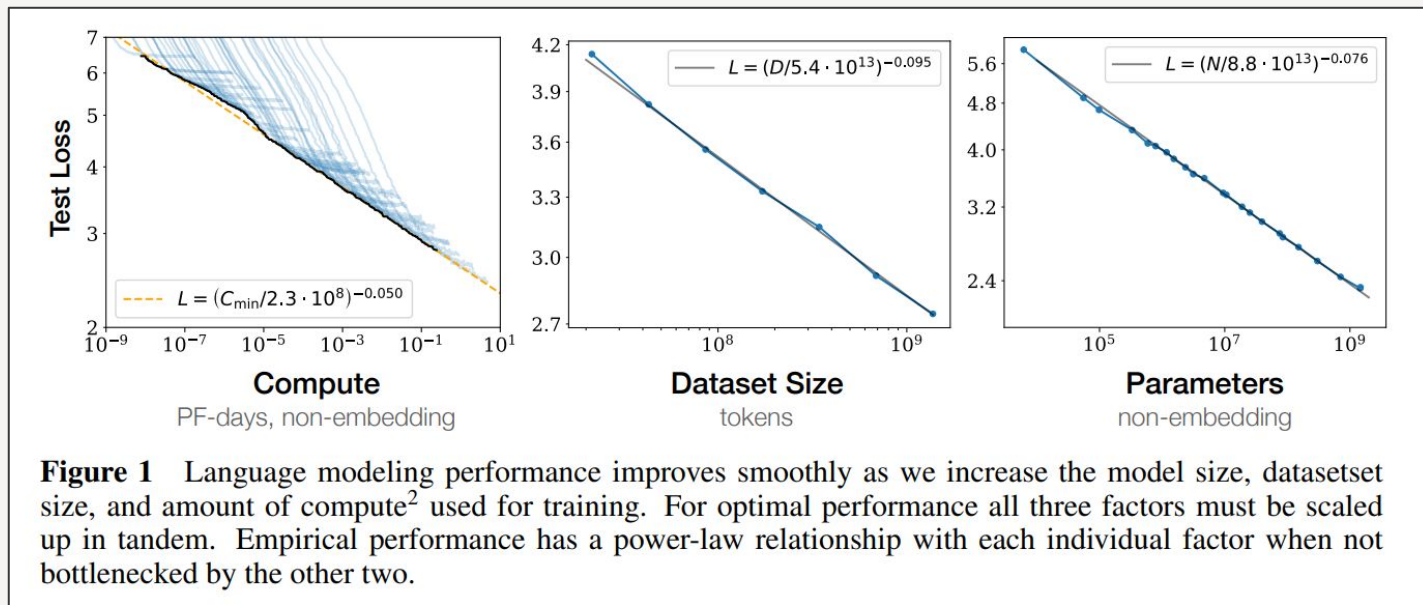
COMPUTE



TEST LOSS

Scaling Laws

- Base: arquitectura vista en clase
- ¿Cómo se escala a los modelos de las empresas de punta?

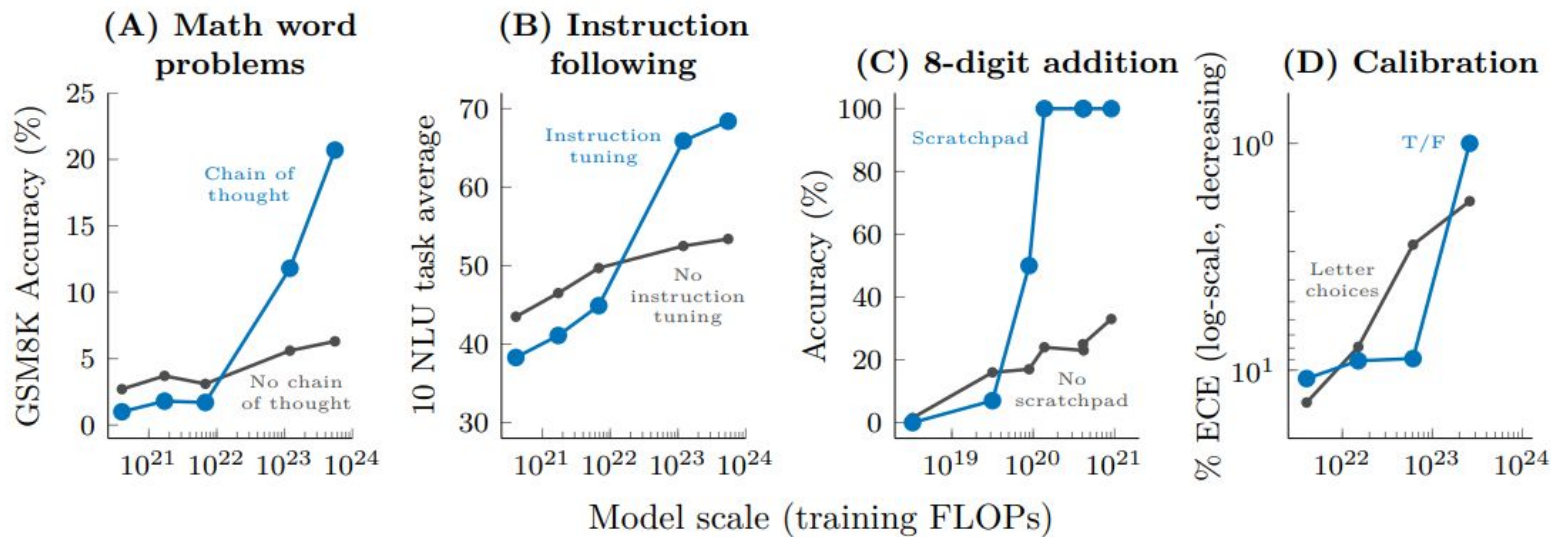


CAPACIDADES GENERALES

Un **modelo de lenguaje** puede...

- Resumir
- Traducir
- Realizar operaciones aritméticas
- Generar código funcional
- Razonamiento de múltiples pasos
- ...

CAPACIDADES EMERGENTES



¿SON REALMENTE CAPACIDADES?

- Evidencia incompleta
- Presión por publicación
- Presión por justificar inversión
- Sesgo que adjudica capacidades al sistema.
- Falta de conocimiento detallado sobre la composición del corpus

(can you explain it as *information retrieval*?)

¿SON REALMENTE CAPACIDADES?

- Evidencia incompleta
- Presión por publicar
- Presión por justificar
- Sesgo que adjudica capacidades al sistema
- Falta de conocimiento detallado sobre la composición del código

(can you explain it as *information retrieval*?)

2407.12220v2 [cs.LG] 30 Oct 2024

Questionable practices in machine learning

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Research
degrees of
freedom

Abstract

Evaluating modern ML models is hard. The strong incentive for researchers and companies to report a state-of-the-art result on some metric often leads to questionable research practices (QRPs): bad practices which fall short of outright research fraud. We describe 44 such practices which can undermine reported results, giving examples where possible. Our list emphasises the evaluation of large language models (LLMs) on public benchmarks. We also discuss “irreproducible research practices”, i.e. decisions that make it difficult or impossible for other researchers to reproduce, build on or audit previous research.

Keywords: machine learning methodology, contamination, cherry picking, misreporting

Table 1: Some questiona

Contamination		Section and description
	Training Contamination	3.1.1: Training on the
	Prompt Contamination	3.1.2: Just putting to (shot)
	RAG Contamination	3.1.3: Leaking benchmarked Generation
	Dirty Paraphrases	3.1.4: Rephrasing test
	Contamination Laundering	3.1.5: Contaminated n
	Thieved Test	3.1.6: Obtaining priva
	User Contamination	3.1.7: Post-training on
	Over-hyping	3.1.8: Tuning hyperpa
	Meta-contamination	3.1.9: Reusing contam
	Semantic duplicates	3.1.10: Train and test

Cherrypicking

	Baseline Nerfing	3.2.1: Optimising train
	Baseline Hacking	3.2.1: Choosing weak
	Runtime Nerfing	3.2.2: Optimising base
	Runtime Hacking	3.2.3: Post-hoc best in
	Benchmark Hacking	3.2.4: Choosing easier benchmarks
	Subset Hacking	3.2.5: Subsetting the benchmark until you win
	Harness Hacking	3.2.6: Choosing part of best evaluation harness



Horace He

@cHHillee

I suspect GPT-4's performance is influenced by data contamination, at least on Codeforces.

Of the easiest problems on Codeforces, it solved 10/10 pre-2021 problems and 0/10 recent problems.

This strongly points to contamination.

1/4

g's Race	implementation, math			greedy, implementation		
and Chocolate	implementation, math			Cat2		
triangle!	brute force, geometry, math			Actions		
	greedy, implementation, math			Interview Problem		
Numbers	brute force			wers		
Line Line	implementation			nd Suffix Array		
or Stairs2	implementation			ther Promotion		
Loves 3 I	math			Forces		
S	implementation, math			d and Append		
	greedy, implementation, sortings			ng Directions		

Evaluation Plausibly

Evaluation No

Evaluation No

Mixture of Experts



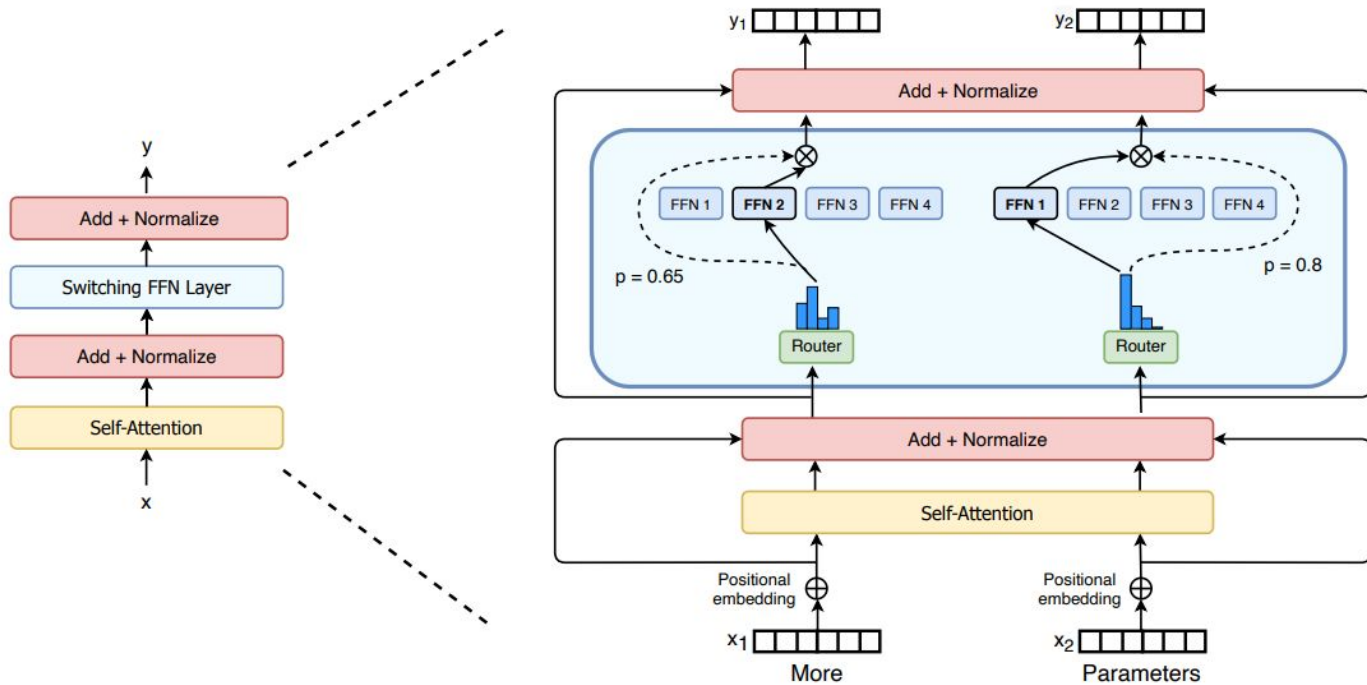
1991: Adaptive Mixtures of Local Experts

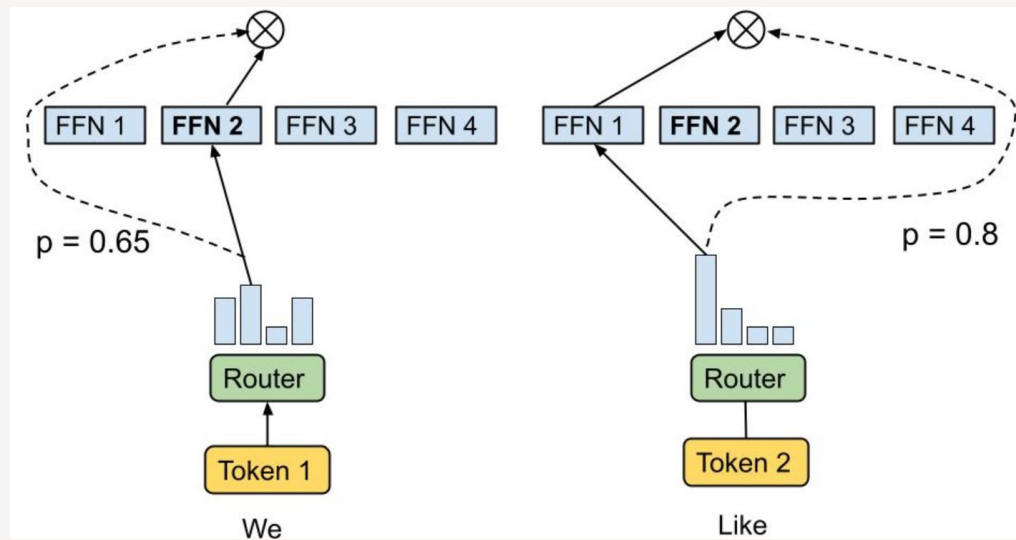


2017: Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer



2021: Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity





$$G(x) = \text{Softmax}(\text{KeepTopK}(H(x), k))$$

$$H(x)_i = x \cdot W_g + \text{ruido}$$

Rotary Positional Encoding

¿Qué busco de un esquema PE?

- Valores únicos en posición de secuencia
- Posición generada según proceso determinístico
- Generalización por fuera del set de entrenamiento

*y que no me ahogue el embedding

¿Qué busco de un esquema PE?

- **Valores únicos en posición de secuencia**
 - Sumarle un entero/Sumarle fracción/Sinusoidal
- **Posición generada según proceso determinístico**
 - Sumarle un entero/~~Sumarle fracción~~/Sinusoidal
- **Generalización por fuera del set de entrenamiento**
 - ~~Sumarle un entero/Sumarle fracción/Sinusoidal~~

*y que no me ahogue el embedding (~~Sumarle un entero~~)

¿Qué busco de un esquema PE?

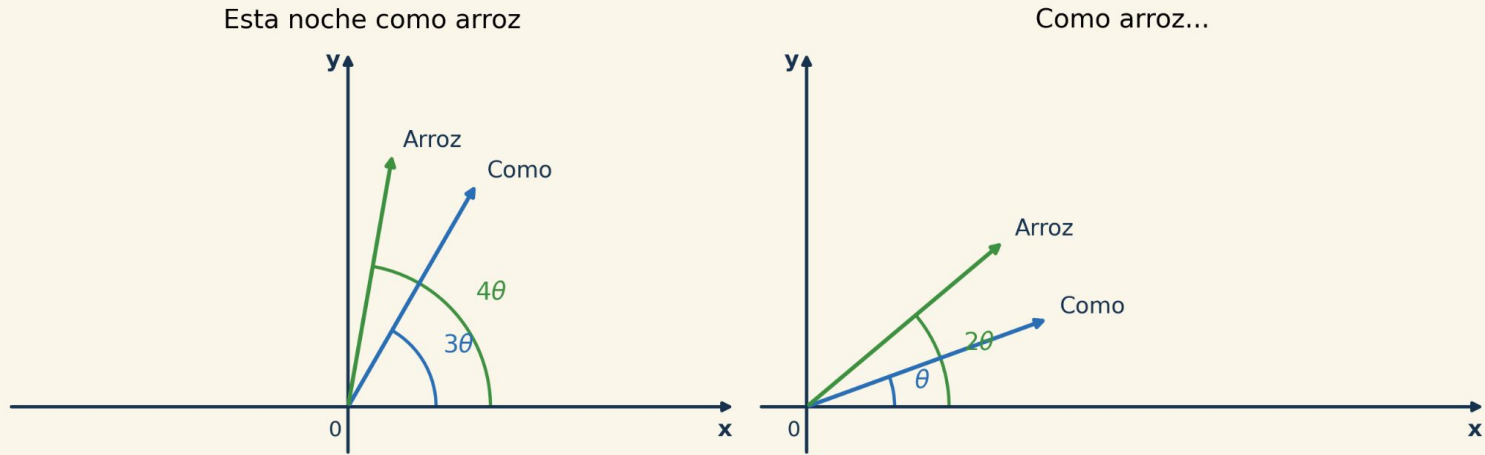
Generalización por fuera del set de entrenamiento

~~○ Sumarle un entero/Sumarle
fracción/Sinusoidal~~

- Mezcla la información del embedding con posicional
- Posiciones absolutas

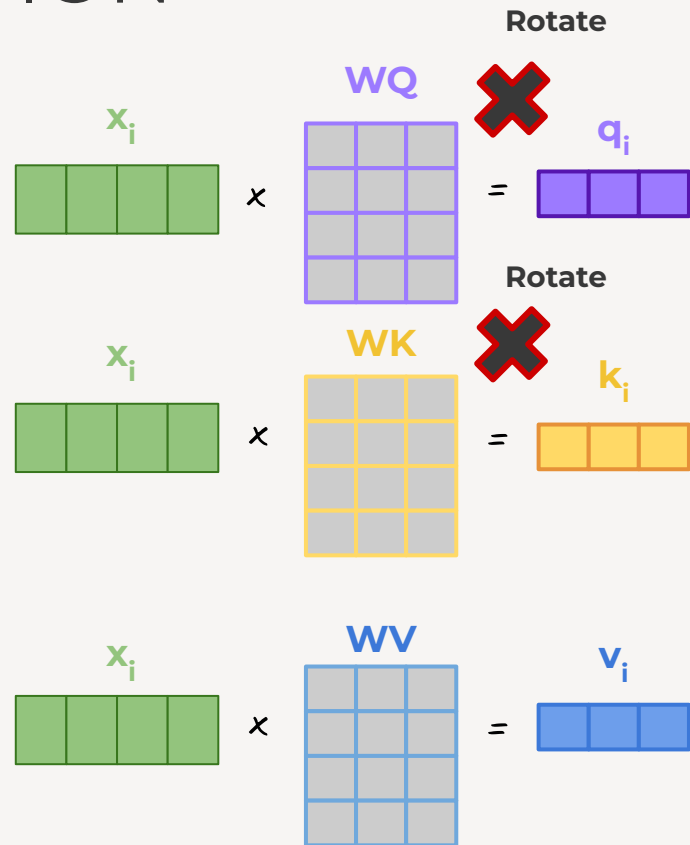
*“Como arroz hoy a la noche”
“Hoy creo que como arroz”*

¿Qué busco de un esquema PE?



CÁLCULO DE SELF-ATTENTION

1. Por cada embedding x_i , se calculan 3 vectores llamados **query**, **key** y **value** multiplicando los embeddings por 3 matrices de pesos (*Attention Weights*) **WQ**, **WK** y **WV**



¿Por qué hacer eso me da relativo?

$$q_{\text{"arroz"}} = R_{2\theta} W_q x_{\text{"arroz"}} \quad k_{\text{"como"}} = R_{\theta} W_k x_{\text{"como"}}$$

$$a_{\text{"c","a"}} = x_{\text{"c"}}^T W_k^T R_{\theta}^T R_{2\theta} W_q x_{\text{"a"}}$$

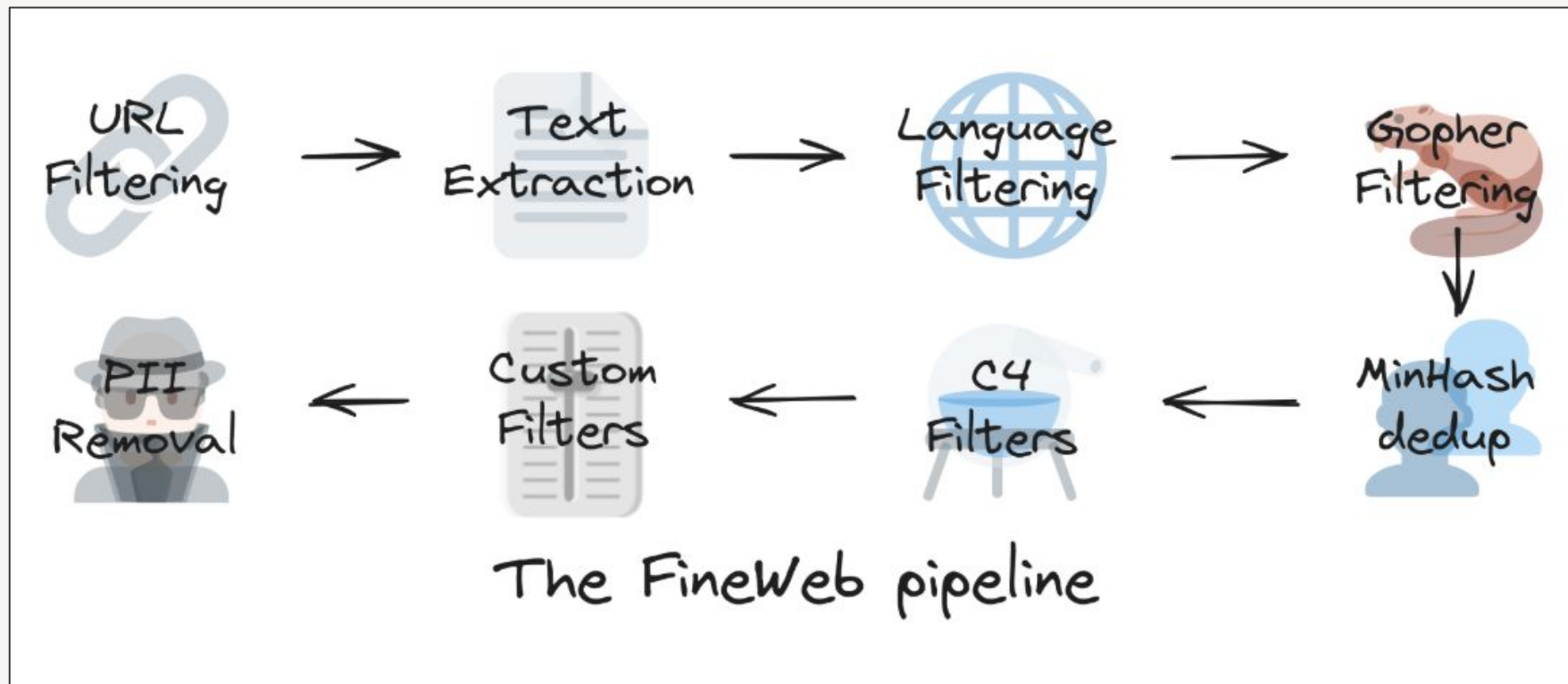
$$a_{\text{"c","a"}} = x_{\text{"c"}}^T W_k^T R_{\theta}^{-1} R_{2\theta} W_q x_{\text{"a"}}$$

$$a_{\text{"c","a"}} = x_{\text{"c"}}^T W_k^T R_{-\theta} R_{2\theta} W_q x_{\text{"a"}}$$

$$a_{\text{"c","a"}} = x_{\text{"c"}}^T W_k^T R_{(2\theta-\theta)} W_q x_{\text{"a"}}$$

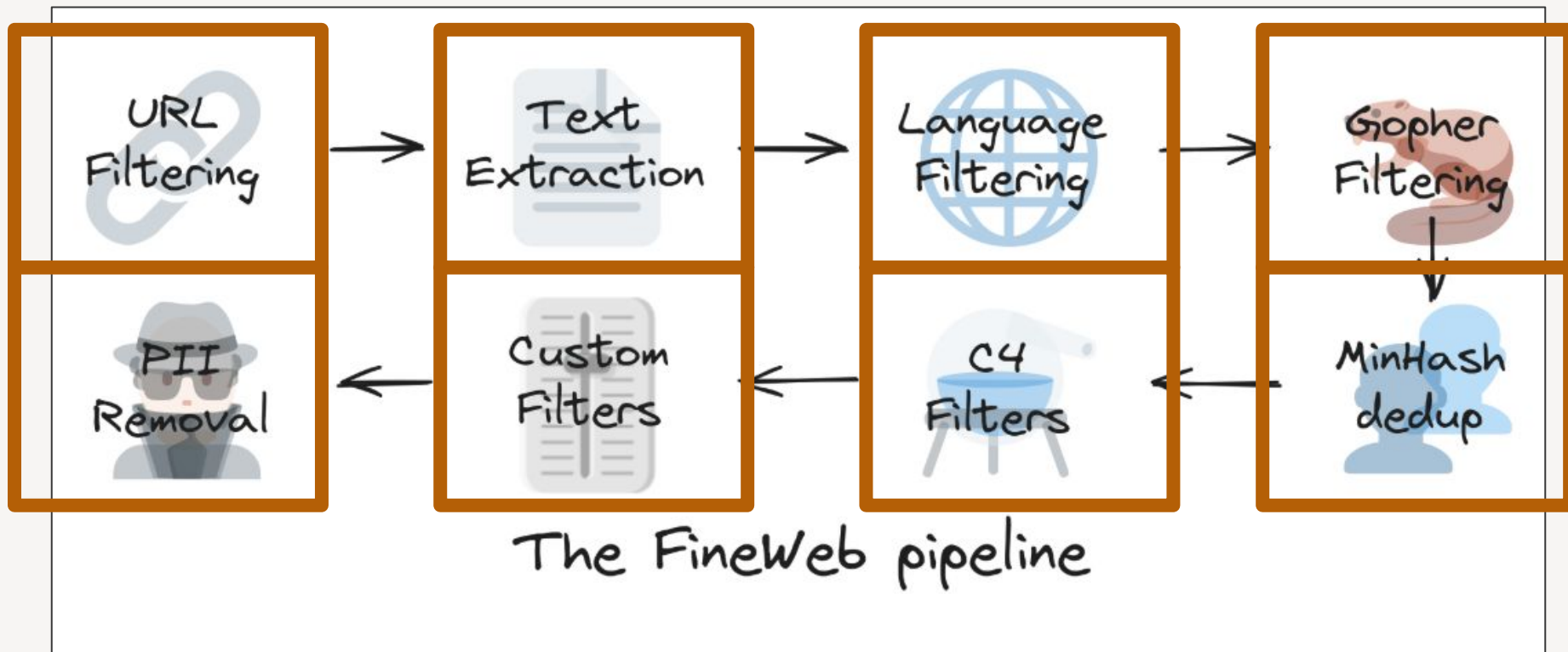
Construcción del corpus

CONSTRUCCIÓN: limpieza y preparación de datos



[FineWeb: decanting the web for the finest text data at scale](#)

CONSTRUCCIÓN: limpieza y preparación de datos



[FineWeb: decanting the web for the finest text data at scale](#)

Avances/Producción

Una (muy) selectiva lista de prácticas:

- ***Scaling Laws***: para configuración del pre-training
- ***Mixture of Experts***: para aprovechar más parámetros a mismo nivel de cómputo + bajar el costo de inferencia
- ***RoPE***: generalización a secuencias largas
- ***Construcción del corpus***: proceso de obtener un corpus de alta calidad
- ***LoRA***: reducir parámetros a fine-tunear
- ***Quantization***: bajar la precisión del modelo

¡ GRACIAS !